

| Paper: 1MA1/1H |         |               |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|---------|---------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question       | Working | Answer        | Mark                                    | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 19             |         | $\frac{2}{5}$ | <p>P1</p> <p>P1</p> <p>P1</p> <p>A1</p> | <p>for first step to solve the problem e.g. <math>\overrightarrow{AC} = -\mathbf{a} + \mathbf{c}</math> or <math>\overrightarrow{OX} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}</math> or demonstrates the location of <math>D</math> and <math>X</math> on the diagram</p> <p>for a correct vector statement using <math>\overrightarrow{CD}</math> eg <math>\overrightarrow{CD} = \overrightarrow{CX} + \overrightarrow{XD}</math> or <math>\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC}</math> or <math>\overrightarrow{OD} = \frac{7}{2}\mathbf{c}</math> or <math>\overrightarrow{CD} = 2.5\mathbf{c}</math> oe</p> <p>for a correct equation or ratio using <math>k</math> eg equating <math>\overrightarrow{XD} = 3\mathbf{c} - \frac{1}{2}\mathbf{a} = \frac{1}{2}(-\mathbf{a} + \mathbf{c}) + \frac{1}{k}\mathbf{c}</math> or <math>\frac{\overrightarrow{OD}}{\overrightarrow{OC}} = \frac{k+1}{k}</math> or <math>k = \frac{1}{2.5}</math> or using a ratio approach eg <math>(\overrightarrow{OC} : \overrightarrow{CD}) = k : 1 = 1 : 2.5</math></p> <p>cao</p> |